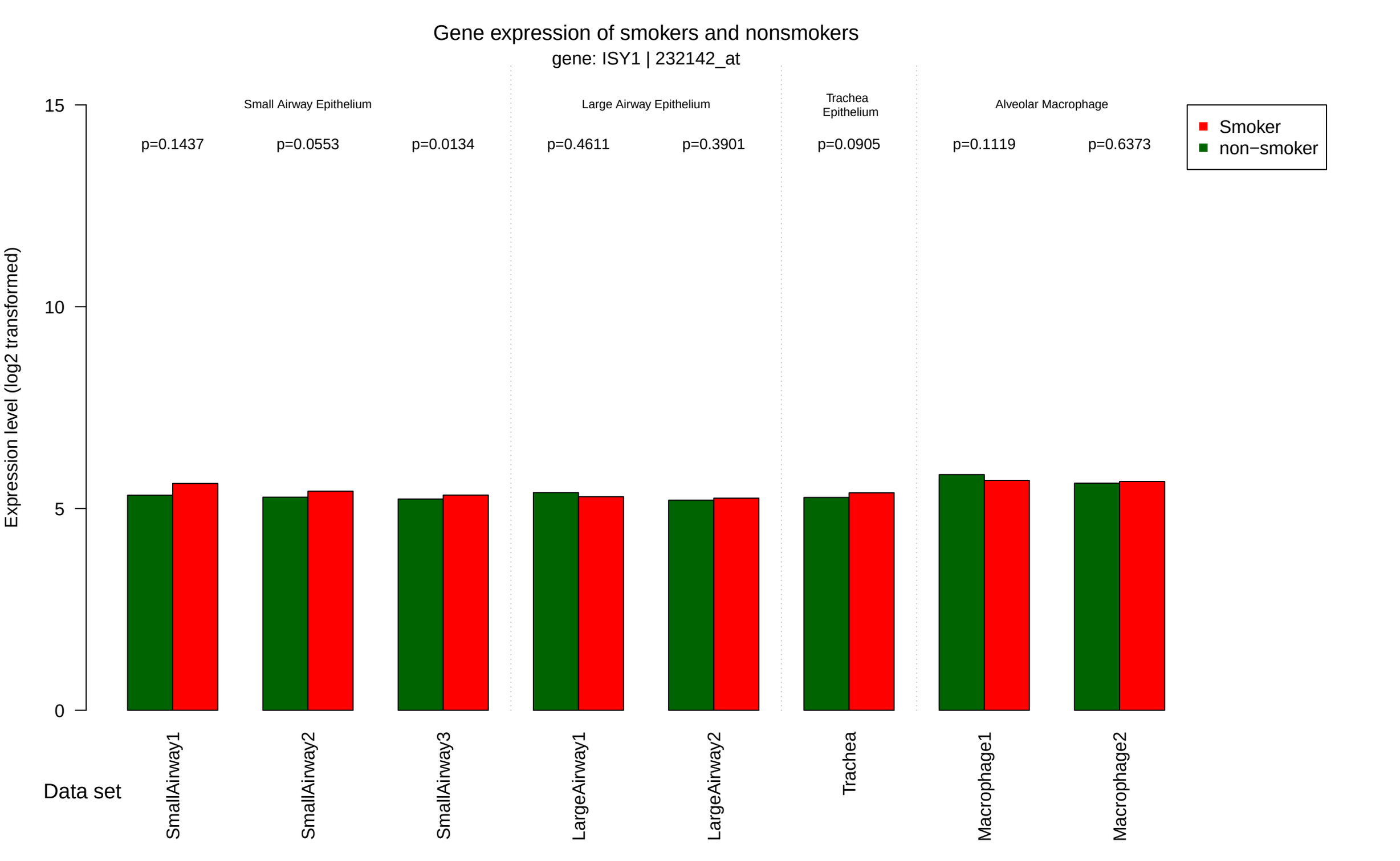
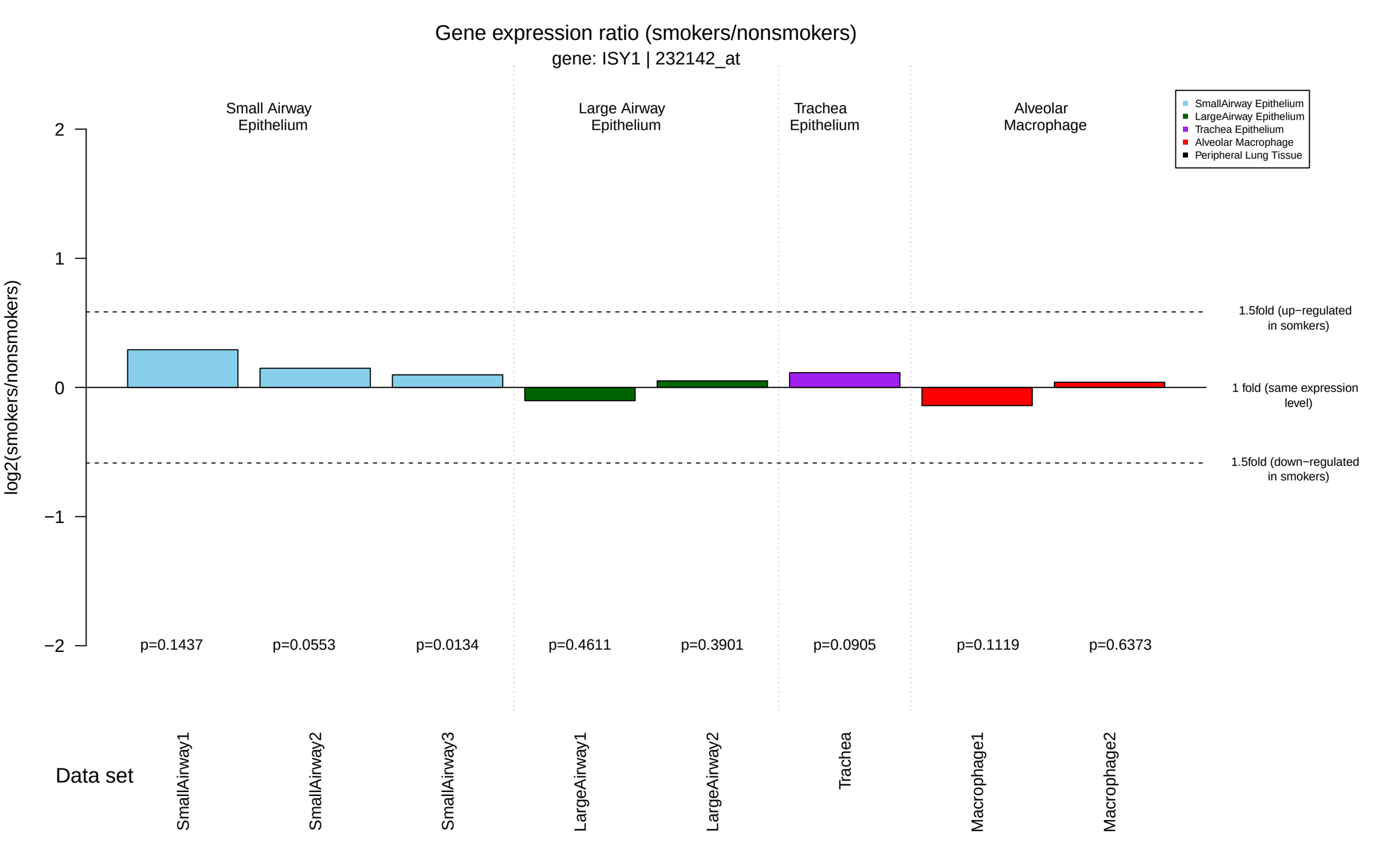
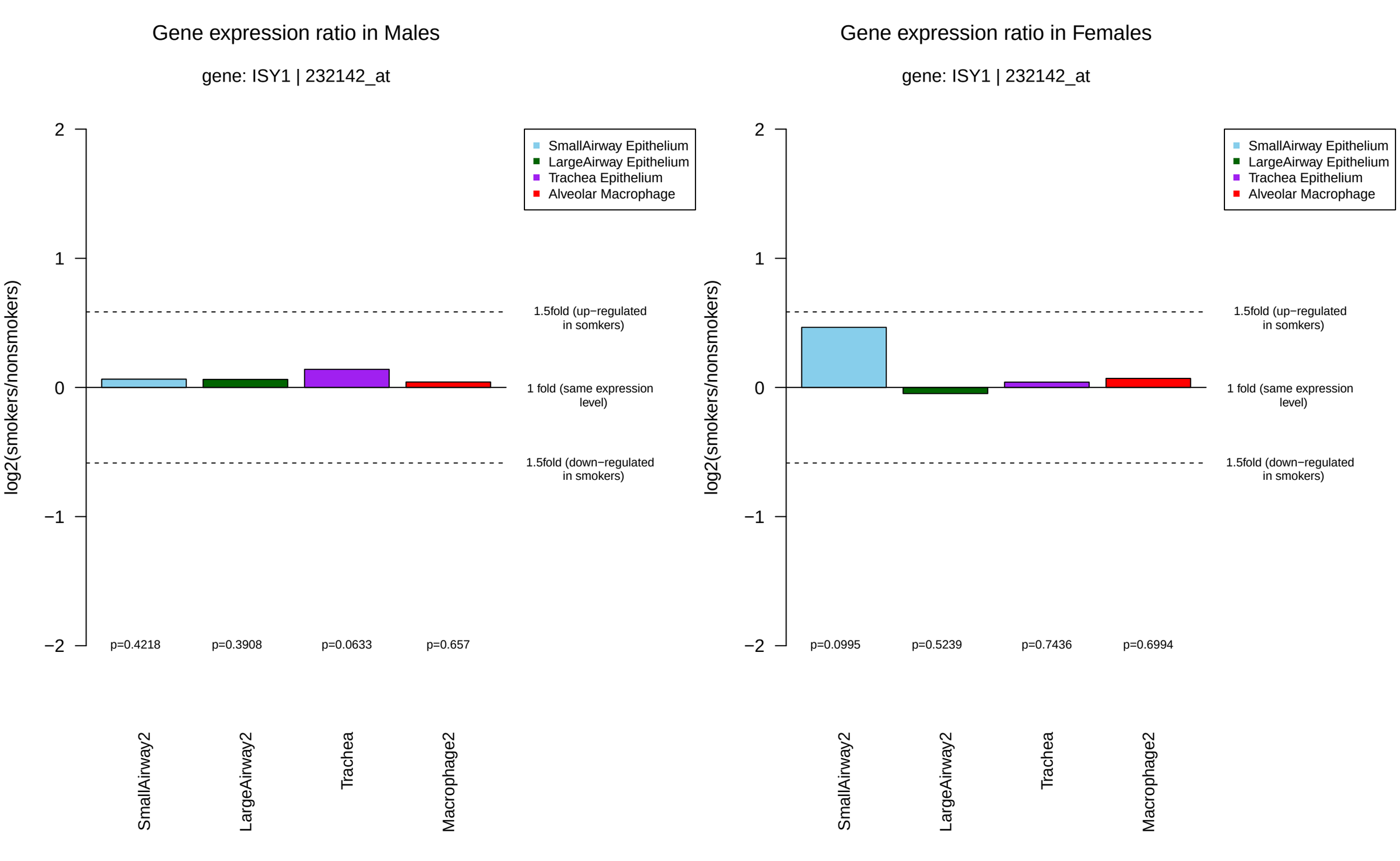
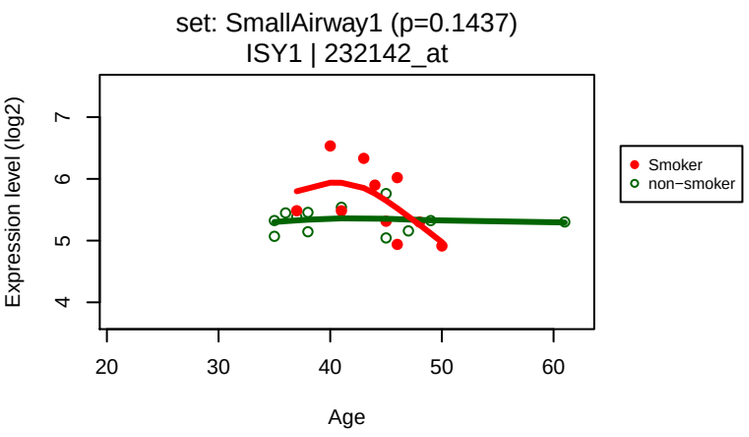
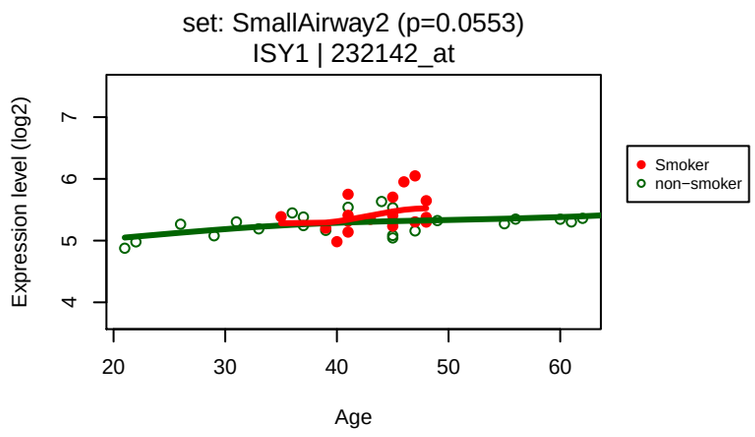
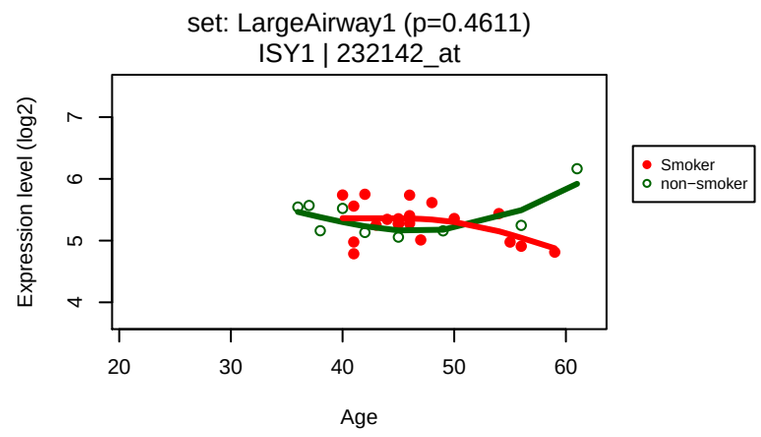

Gene expression of smokers and nonsmokers
gene: ISY1 | 232142_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1437
p=0.0553
p=0.0134
p=0.4611
p=0.3901
p=0.0905
p=0.1119
p=0.6373
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

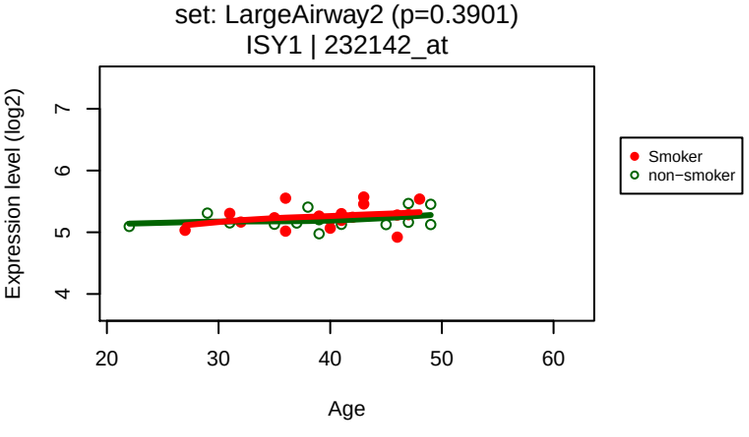

Gene expression ratio (smokers/nonsmokers)
gene: ISY1 | 232142_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1437
p=0.0553
p=0.0134
p=0.4611
p=0.3901
p=0.0905
p=0.1119
p=0.6373
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

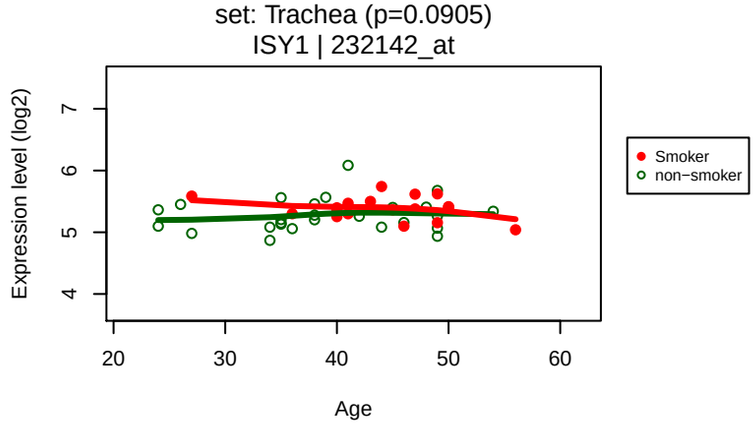

Gene expression ratio in Males
gene: ISY1 | 232142_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4218
p=0.3908
p=0.0633
p=0.657
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISY1 | 232142_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0995
p=0.5239
p=0.7436
p=0.6994
SmallAirway2
LargeAirway2
Trachea
Macrophage2

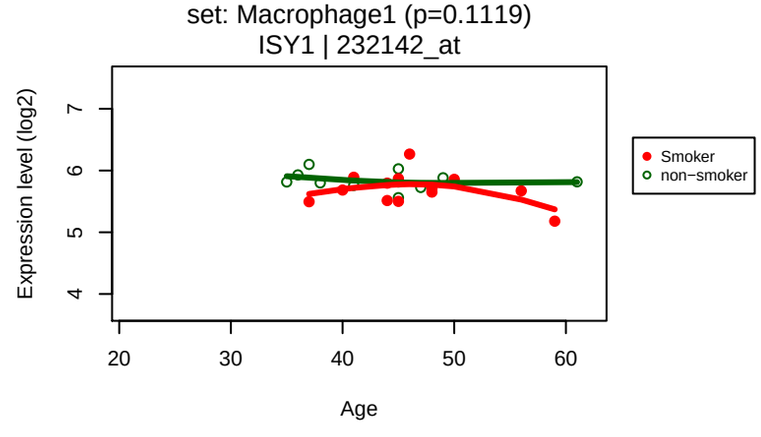

set: SmallAirway1 (p=0.1437)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker

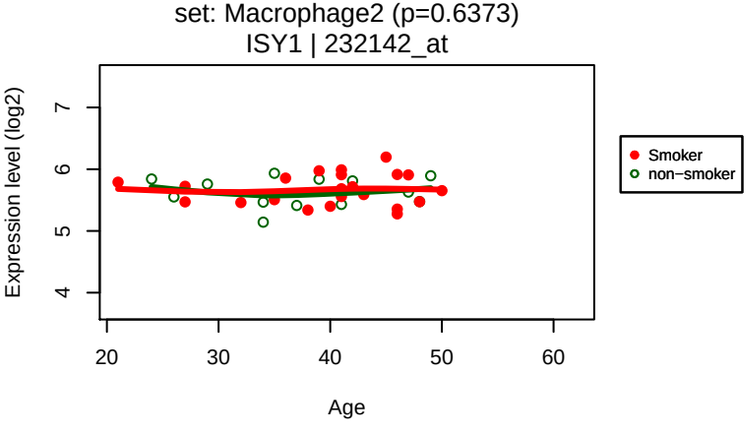

set: SmallAirway2 (p=0.0553)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker

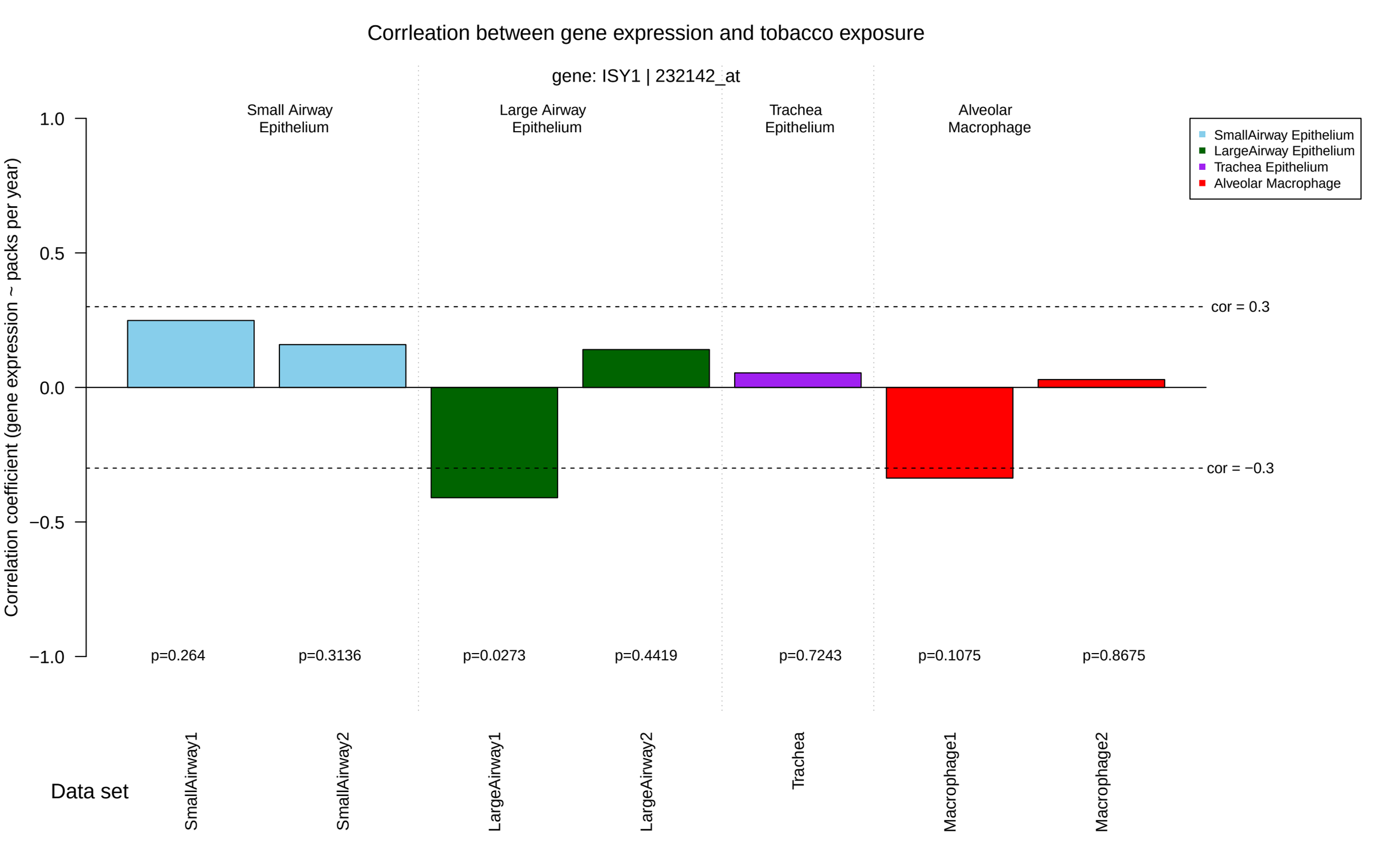

set: LargeAirway1 (p=0.4611)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3901)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0905)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1119)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6373)
ISY1 | 232142_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISY1 | 232142_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.264
p=0.3136
p=0.0273
p=0.4419
p=0.7243
p=0.1075
p=0.8675
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2